Expanding Attention Lecture Notes

Sunday, October 26, 2025 6:36 PM

Makeup test this weekend

Problem with attention: quadratic growth O(n^2)
- Double n and the complexity quadruples
- Air resistance is quadratic
- Polynomial growth not exponential

If we have more context then we can ask bigger questions

Infer end of book from start, need attention to whole book
- Maybe don't need everything, can leave some out

Only have to recompute the next word
- Only computing a linear amount more for each token
- But it does cost more and more each time
- Overall cost at the end ends up being n^2

Goal is linear overall, not linear per token
- Cost per token would be constant

BigBird method:
- Global - O(n)
  - Can see and attend everything
- Local - O(n)
  - Can see and attend neighbors
- Random - O(n)
  - Random tokens that attend to one another
  - Helpful because can attach one node, which goes to another, and so on
- Sums to a near complete graph while having better computational complexity - O(n)

- 1 of many alternative methods to attention that helps us scale

Single instruction, multiple data
- Takes advantage of hardware - GPU/TPU

Cost to train GPT2 = $256 per hour

Quantization
- Model compression method
- Train with smaller bit sizes
- Take existing model and round weights down
- Gives you more vram free

Prompt Compression
- Pass prompt to other model to compress it
- Slim down prompt to keep context smaller

Multi-query and group attention

- Computing a single query and key that gets passed to multiple attention heads
- Size 2048x2048
- Compress data of tokens and use compressed version for keys and queries

Cutting edge, don't know what will emerge as best

What to know:

- Quantization
- Prompt Compression
- Sparse pattern
- Quadratic growth - O(n^2)